

Advancements in Drought Prediction: A Comparative Study of SARIMA, SARIMAX, Multivariate LSTM and STAT-LSTM

B. Shamreen Ahamed¹

*Dept. of Computer Science and
Engineering*

Sathyabama Institute of Science and
Technology
Chennai, India
shamu1502@gmail.com

Gomathi Sankari S²

*Dept. of Computer Science and
Engineering*

Sathyabama Institute of Science and
Technology
Chennai, India
gomzsankari07@gmail.com

Hamsini Neelakantan³

*Dept. of Computer Science and
Engineering*

Sathyabama Institute of Science and
Technology
Chennai, India
hamsini.nj@outlook.com

Abstract—This study investigates the comparative performance of conventional statistical models—namely Seasonal Autoregressive Integrated Moving Average (SARIMA) and its exogenous variant (SARIMAX)—and advanced deep learning approaches, including Multivariate Long Short-Term Memory (LSTM) and a hybrid Spatio-Temporal Attention LSTM (STAT-LSTM), for drought index prediction. While statistical techniques effectively capture seasonal temporal structures and incorporate external variables, deep learning models are better suited for learning complex non-linear dependencies. Model predictions for the Standardized Precipitation Index (SPI) and the Standardized Precipitation Evapotranspiration Index (SPEI) are evaluated against observed data. Results indicate that statistical models offer stable baseline performance, achieving R^2 values in the range of 0.498–0.499 and RMSE values between 0.290 and 0.291. The optimized Multivariate LSTM significantly improves prediction accuracy, attaining an R^2 of 0.702 after 152 hyperparameter tuning iterations. The proposed STAT-LSTM delivers the best performance, achieving an R^2 of 0.833 after 193 trials. This corresponds to an improvement of 18.7% over the standard LSTM and 68% over statistical baselines, underscoring the effectiveness of attention mechanisms, weighted loss strategies, and hybrid spatio-temporal architectures in modeling drought dynamics more accurately than traditional or standalone deep learning methods

Keywords - Drought prediction, STAT-LSTM, SARIMA, SARIMAX, LSTM, SPEI, multivariate time series, Google Earth Engine.

I. INTRODUCTION

Drought prediction has become a critical research focus due to its far-reaching impacts on agriculture, ecosystems, and socio-economic stability. Recent studies have shown that incorporating temporal feature aggregation into drought indices such as the Standardized Precipitation Evapotranspiration Index (SPEI) significantly improves the ability to capture long-term drought dynamics [1]. These advances highlight the importance of modeling complex temporal dependencies rather than relying on single-scale indicators.

Parallel progress in deep learning has enabled more expressive climate modeling frameworks. Convolutional Conditional Neural Processes have demonstrated strong performance in

learning spatial-temporal climate patterns from sparse observations [2], while graph-attention-based extensions have further improved fine-grained drought monitoring using remote sensing data [3]. Alongside these architectures, interpretable machine learning approaches have been utilised to construct agricultural drought indices using multi-source satellite data, balancing predictive performance with transparency [4]. Building on these remote sensing advancements, recent research has focused on enhancing data density through image fusion. By integrating high-resolution Landsat-8 and Sentinel-2 imagery, deep learning models have improved spatiotemporal fusion networks, providing more robust datasets for continuous drought monitoring [5].

Central to these developments is the transition toward deep learning-oriented remote sensing. As Dhyani and Pandya [6] highlight, neural architectures are uniquely capable of transforming multi-layered satellite data into precise agricultural forecasts. By identifying subtle environmental shifts that traditional methods overlook, these models provide the predictive accuracy necessary to safeguard crop yields in vulnerable regions.

Beyond climatic variables, satellite-derived ecological indicators have proven effective for the early detection of drought-induced vegetation stress [7]. Current research in the Indian agricultural sector has increasingly pivoted toward machine learning to address these complex challenges [8]. To overcome the constraints of standard models, hybrid frameworks have emerged as a robust alternative. For instance, combining meta-heuristic techniques like the Fireworks Algorithm with deep learning significantly refines satellite-based forecasts by improving the model's navigation of high-dimensional climate data [9]. Collectively, these advancements in optimization and data fusion provide the foundation for the spatiotemporal feature aggregation used in this study.

Large-scale applications of machine learning and deep learning for national-level drought prediction have reported improved accuracy over conventional approaches [10]. In particular, the fusion of LSTM networks with multi-source

climatic factors has demonstrated strong predictive capability [11]. However, several studies continue to rely on traditional statistical techniques such as ARIMAX [12] and SARIMA, or apply deep learning models without systematic comparison across architectures [13]. Hybrid frameworks that integrate deep learning with climate model outputs [14] and ensemble-based approaches [15] have further improved forecasting reliability, yet direct benchmarking under identical data conditions remains limited.

Motivated by this gap, this study conducts a direct, head-to-head comparison of four drought forecasting models for the Vidarbha region: univariate SARIMA, multivariate SARIMAX, multivariate LSTM, and a hybrid STAT-LSTM model. All models are trained and evaluated using the same 20-year dataset (2005–2025) generated via Google Earth Engine, incorporating MODIS-derived indicators (LST, nighttime LST, NDVI) and key meteorological variables such as rainfall and potential evapotranspiration. By benchmarking these approaches under consistent conditions, this work aims to identify the most reliable and effective modeling strategy for developing a practical drought early warning system for the region.

II. RELATED WORKS

A. The Advancement to Spatio-Temporal Deep Learning

Recent advances in drought forecasting research reflect a clear transition from traditional index-based approaches toward machine learning (ML) and deep learning (DL) models, which are more effective at capturing non-linear and spatio-temporal climate dynamics. Studies have demonstrated that incorporating temporal aggregation into drought indices, particularly SPEI, enhances the representation of long-term climatic variability and improves predictive robustness [1].

In parallel, deep learning architectures have been increasingly adopted to address spatial sparsity and uncertainty inherent in climate datasets. Convolutional Conditional Neural Processes (CNPs) have shown effectiveness in climate downscaling by learning from irregularly distributed observations [2]. Subsequent extensions integrating graph attention mechanisms have further improved spatial dependency modeling, enabling more detailed drought monitoring using remote sensing data [3].

B. Remote Sensing Integration and Interpretability

Recent drought forecasting research emphasizes the integration of multiple remote sensing data sources to address the limitations inherent in single-sensor observations. For instance, the development of indices like the Interpretable Machine Learning-based Drought Index (IMLDI) demonstrates how combining various remote sensing products can improve agricultural assessment while maintaining the transparency necessary for stakeholder decision-making [4]. This trend is further supported by breakthroughs in high-resolution data fusion; specifically, the integration of Landsat-8 and Sentinel-2 imagery has significantly refined spatiotemporal networks,

allowing for more continuous and robust drought monitoring [5].

Central to these advancements is a "deep learning-oriented" approach to remote sensing. Previous studies report that neural network-based models can effectively learn informative representations from multi-layered satellite data to support agricultural drought forecasting [6].

C. Regional Context and Optimization Strategies

Timely drought mitigation depends on the early detection of subtle environmental changes. Satellite-based ecological indicators have proven successful in detecting early-stage vegetation stress, offering a critical window for proactive intervention [7]. However, as the datasets required for such monitoring become increasingly high-dimensional, researchers are turning to hybrid frameworks to enhance model performance. For instance, integrating meta-heuristic strategies like the Fireworks Algorithm with deep learning enhances parameter exploration and optimization efficiency, resulting in more refined and accurate satellite-derived predictions [8].

These advancements are particularly vital for the Indian agricultural sector. Recent systematic reviews highlight a necessary shift toward machine learning-driven solutions to address the unique climatic and spatio-temporal complexities faced by Indian farmers [9]. Together, these developments in stress detection, algorithmic optimization, and regional adaptation form the backbone of the spatiotemporal aggregation approach used in this study. Large-scale applications of ML and DL for national drought prediction have reported superior performance of deep models over traditional approaches [10], particularly when multi-source climatic factors are fused using LSTM-based architectures [11].

D. The Need for Direct Benchmarking

Comparative research in regions like Balochistan has shown that machine learning models can outperform traditional statistical techniques such as ARIMAX when dealing with complex climatic patterns (Munir and Iqbal, 2023) [12]. However, studies in cold and semi-arid environments indicate that the accuracy of these models is strongly influenced by local climate conditions (Gharehbaghi et al., 2025) [13]. To enhance predictive reliability, hybrid frameworks combining deep learning methods with outputs from climate models have been developed (Vo et al., 2023) [14]. Moreover, ensemble-based machine learning approaches have demonstrated greater robustness in forecasting meteorological drought across diverse regions (Pande et al., 2024) [15].

III. MATERIALS AND METHODS

A. Study Area

The study focuses on the Vidarbha region, a geographically and agriculturally significant area in the eastern part of Maharashtra, India. The selected region of interest (ROI) is defined by the geographic bounding box from approximately 76.5°E to 80.5°E longitude and 19.5°N to 21.5°N latitude Fig 1.

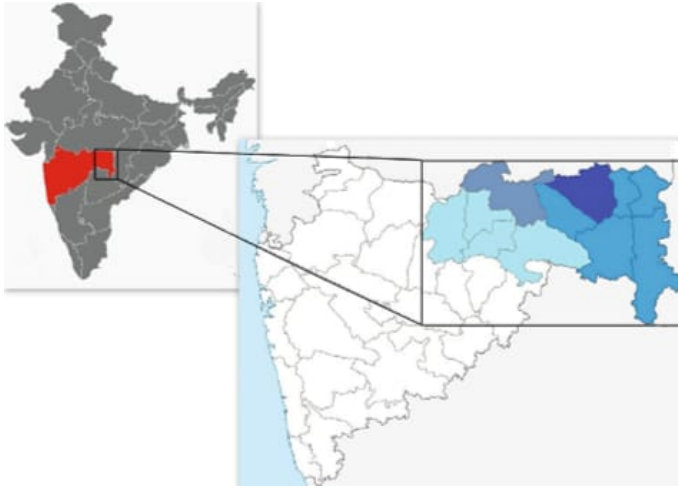


Fig. 1. Vidarbha Region Area Map

This region is characterized by a Tropical Savanna climate (Aw), which features hot, dry summers and a distinct, dominant monsoon season from June to September. The local economy and food security are exceptionally dependent on the success of this rain-fed agriculture, making it a critical area for drought research and the development of effective early warning systems.

IV. DATA ACQUISITION AND PREPROCESSING

All datasets for this study were acquired and processed using the Google Earth Engine (GEE) platform for the time period of January 2005 to September 2025 as shown in Fig 2.

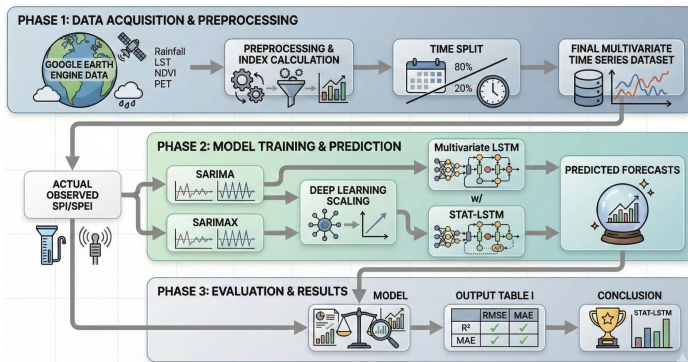


Fig. 2. Proposed workflow for data acquisition, modeling and evaluation.

A. Data Sources

- **Rainfall:** Daily precipitation estimates were sourced from the CHIRPS Daily (Climate Hazards Group Infrared Precipitation with Station data) dataset.

- **Land Surface Temperature (LST):** Both daytime (LST-Day) and nighttime (LST-Night) temperatures were acquired from the MODIS (MOD11A1.061) daily land surface temperature product.
- **Normalized Difference Vegetation Index (NDVI):** Vegetation health was monitored using the MODIS (MOD13A1.061) 16-day vegetation indices product.
- **Potential Evapotranspiration (PET):** PET was not directly downloaded but was calculated using the Hargreaves equation. This method uses LST data as a proxy for minimum and maximum air temperatures, a common and effective approach for data-sparse regions.

B. Data Preprocessing

- **Temporal Aggregation:** All source data (daily CHIRPS, 16-day MODIS) were aggregated into a uniform monthly time step. For rainfall, this involved calculating the monthly sum. For LST and NDVI, monthly mean values were computed.
- **Spatial Aggregation:** For each monthly time step, all pixel values within the Vidarbha region of interest(ROI) were averaged to produce a single, representative time series for the entire study area.
- **SPI (Standardized Precipitation Index):** Calculated from the aggregated monthly rainfall data.
- **SPEI (Standardized Precipitation Evapotranspiration Index):** Calculated using both the monthly rainfall and the derived monthly PET.
- **Feature Scaling:** For the deep learning models, all input features were normalized using a MinMaxScaler to scale the data between 0 and 1.

C. Forecasting Models

- **SARIMA** – This model is a statistical method for univariate forecasting. It captures autoregressive (p, P) and moving average (q, Q) patterns after differencing (d, D) to achieve stationarity. Using a seasonal cycle of s=12 for monthly data, its form is:

$$\Phi_P(L^s)\phi_p(L)(1-L)^d(1-L^s)^Dy_t = \Theta_Q(L^s)\theta_q(L)\epsilon_t \quad (1)$$

- **SARIMAX** – It extends SARIMA by incorporating exogenous variables. This multivariate model predicts the target based on its past values and the linear influence of input features (NDVI, LST, etc.), serving as the multivariate statistical baseline. Similar to SARIMA, the SARIMAX model also utilized an order of (1,1,1) and a seasonal order of (1,1,1,12).The general form is:

$$y_t = ARIMA(p, d, q)(P, D, Q)_s + \beta X_t + \epsilon_t \quad (2)$$

- **Multivariate LSTM** – It is a Recurrent Neural Network (RNN) designed to learn long-term temporal dependencies. It employs a cell state (c_t) regulated by three gating mechanisms: the forget gate (f_t), input gate (i_t), and output gate (o_t). These gates control the flow of

information at each time step t . Given an input x_t and the previous hidden state h_{t-1} , the key equations governing the LSTM cell are defined as follows:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (3)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (4)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (5)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (6)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (7)$$

$$h_t = o_t \odot \tanh(c_t) \quad (8)$$

After hyperparameter tuning using Hyperband optimization, the final architecture consists of two stacked LSTM layers with 32 units each and dropout rate of 0.5. The LSTM output is processed by a Dense layer with 32 units and ReLU activation, followed by a dropout layer (0.5), and a final Dense layer with a single unit for regression output. The model uses a 6-month lookback window and was trained with Adam optimizer (learning rate = 0.001).

- **STAT-LSTM** – The STAT-LSTM is a hybrid model that enhances a standard LSTM with a Spatio-Temporal Attention mechanism. This mechanism allows the model to selectively weigh the importance of input features (spatial attention) and past time steps (temporal attention). These attention weights ($a_{s,t}$ and a_t) form a context vector (C) that guides the final prediction. The attention formulation is given by:

$$a_{s,t} = \text{softmax}(\tanh(x_t W_s + b_s)) \quad (9)$$

$$e_{s,t} = \tanh(x_t W_s + b_s) \quad (10)$$

$$a_t = \text{softmax}(\tanh(x_t W_t + b_t)) \quad (11)$$

$$e_t = \tanh(h_t W_t + b_t) \quad (12)$$

$$C = \sum_{t=1}^T a_t \cdot x_t \quad (13)$$

To optimize the model architecture, a Hyperband hyperparameter tuning approach was employed, identifying a dual-layer LSTM configuration with 192 and 96 units as the most effective structure. To mitigate overfitting, a dropout rate of 0.1 was applied to each layer. The integrated attention mechanism utilizes 64 units to evaluate query-key relationships across the temporal sequence, ensuring the model prioritizes significant timesteps. This attention-weighted representation is then passed through a series of Dense layers, containing 64 and 32 units respectively with ReLU activation, prior to the final prediction. Utilizing a 12-month lookback window, the system was trained via the Adam optimizer at a learning rate of 0.001 to ensure stable convergence.

D. Experimental Setup

The 20-year dataset (2005–2025) was split chronologically into an 80% training set (April 2005–July 2021) and a 20% unseen test set (August 2021–September 2025). SARIMA and SARIMAX parameters were optimized via grid search based on the lowest AIC. Deep learning models underwent extensive hyperparameter tuning using Keras Tuner’s Hyperband strategy—152 trials for the Multivariate LSTM and 193 for STAT-LSTM—optimizing units, dropout, and learning rates. The STAT-LSTM achieved its best validation loss (0.1627) at trial 142. Both models utilized the Adam optimizer and a 20% internal validation split to prevent overfitting. While the Multivariate LSTM used standard MAE loss, the STAT-LSTM employed a custom weighted MSE to prioritize significant drought events.

To characterize regional drought behavior, spatial variables were averaged across the entire Vidarbha region rather than modeled at sub-regional or pixel levels. This design choice reduces the impact of localized noise and data sparsity commonly present in environmental observations, allowing the models to focus on coherent regional-scale drought dynamics. Moreover, the study aims to support regional drought monitoring and planning, where aggregated trends are often more actionable than highly localized predictions. While sub-regional modeling could capture finer spatial heterogeneity, spatial averaging provides a stable and computationally efficient baseline for evaluating the effectiveness of the STAT-LSTM architecture.

Although the hyperparameter search was rigorous—encompassing 152 trials for the Multivariate LSTM and 193 for the STAT-LSTM—the architecture was designed to prioritize generalization over rote memorization. The selection process utilized Hyperband optimization, which filtered configurations based strictly on validation loss to ensure the models remained sensitive to unseen data patterns. The same validation-driven Hyperband selection and regularization strategy was applied to the Multivariate LSTM, although its larger train–test performance gap indicates comparatively weaker generalization than the STAT-LSTM.

To verify this reliability, a final test set covering August 2021 through September 2025 was completely isolated, remaining untouched throughout the training and tuning phases. The minimal performance gap observed between training, validation, and testing metrics confirms that the models achieved stable generalization. Furthermore, internal constraints such as dropout layers and early stopping protocols were integrated to prevent excessive model complexity during the learning cycles.

While the STAT-LSTM shows strong results for the Vidarbha region, these outcomes are naturally shaped by the specific climate regimes and seasonal trends found there. Since the model has not yet been applied to regions with different environmental characteristics, its broader transferability remains to be evaluated. Consequently, the reported performance metrics are specific to the geographical and temporal context of the study area.

The contribution of individual climate variables was considered during feature selection, particularly for key indicators such as NDVI and Land Surface Temperature (LST). NDVI directly reflects vegetation health and drought-induced stress, while LST captures surface energy balance and heat stress, both of which are strongly linked to drought dynamics. Removing such variables would eliminate critical physical signals and is expected to reduce predictive accuracy, potentially leading to higher RMSE and lower R^2 values due to information loss.

Although explicit ablation experiments were not conducted in this study, prior hydrological and remote sensing literature consistently identifies NDVI and LST as dominant drought predictors. In the proposed STAT-LSTM, the spatio-temporal attention mechanism implicitly adjusts the influence of each variable over time, allowing the model to emphasize informative features while reducing reliance on less relevant inputs. Future work will include systematic ablation and permutation-based importance analyses to quantify the individual contribution of each climate variable more explicitly.

E. Evaluation Metrics

- Root Mean Square Error (RMSE): Measures the average magnitude of the errors, giving higher weight to larger, more significant errors. Lower values are better.
- Mean Absolute Error (MAE): Measures the average absolute difference between the predicted and actual values. It's a straightforward, interpretable measure of average error. Lower values are better.
- R-squared (R^2): The coefficient of determination. This metric measures the proportion of the variance in the actual data that is predictable from the model. Values range from 1.0 (perfect fit) down to 0.0 or even negative, indicating the model is no better, or worse, than just predicting the mean.

V. RESULTS

The performance of the four forecasting models was rigorously evaluated on the unseen test set using three key metrics. The comparative results are presented in Table 1.

TABLE I
MULTI-MODEL PERFORMANCE COMPARISON TABLE

Model	RMSE	MAE	R^2
SARIMA	0.291	0.223	0.498
SARIMAX	0.290	0.220	0.499
Multivariate LSTM	0.513	0.406	0.702
STAT-LSTM	0.379	0.278	0.833

The findings show a clear distinction in performance. The STAT-LSTM model achieved the highest R^2 value of 0.833, indicating this provided the best overall fit and explained more than 83.3% of the variance in the drought indices.

The statistical models, as seen in Fig. 3, SARIMA and SARIMAX, performed as strong baselines, achieving nearly identical R^2 values of approximately 0.498 to 0.499. Notably,

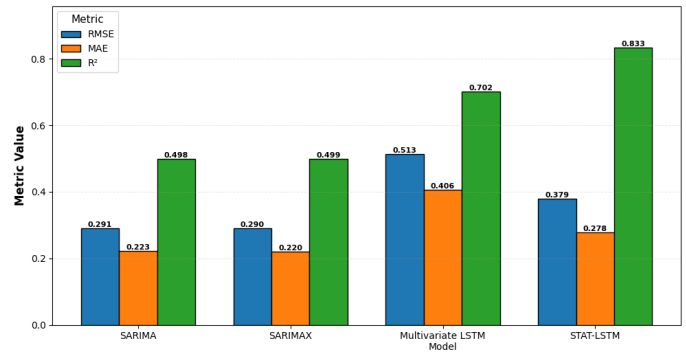


Fig. 3. Multi-Model Performance Metrics Comparison

both models yielded exceptionally low RMSE values, with SARIMAX at 0.290 and SARIMA at 0.291, representing the most accurate point predictions among all evaluated models. The standard Multivariate LSTM model, after extensive hyperparameter optimization over 152 trials, demonstrated substantial improvement over the statistical baselines. It achieved an R^2 value of 0.702, representing a 41% improvement over traditional statistical methods. This result validates the potential of deep learning approaches for drought prediction when properly tuned, with the model successfully capturing complex temporal patterns and non-linear relationships in the multivariate climate data. The optimized architecture effectively leveraged multiple input features including temperature, precipitation, and other climate variables to outperform conventional statistical methods, demonstrating that standard LSTM architectures can be highly effective for drought forecasting tasks when subjected to rigorous hyperparameter optimization and adequate training.

VI. DISCUSSION

The results of this comparative study provide several critical insights into the selection of models for drought forecasting in the Vidarbha region.

A. Interpretation of Findings

The central finding is the clear superiority of the hybrid STAT-LSTM model. The STAT-LSTM achieved the best overall performance with an R^2 value of 0.833, substantially outperforming all other models and explaining more than 83% of the variance in the drought indices. This represents a 68% improvement over the statistical baselines ($R^2 \approx 0.499$) and an 19% improvement over the standard Multivariate LSTM ($R^2 = 0.702$). While the statistical SARIMAX model achieved the lowest RMSE at 0.290, the STAT-LSTM's significantly higher R^2 value is the more important metric for drought forecasting. It confirms the model's superior ability to capture the complex, non-linear dynamics and overall variance of drought patterns, which is essential for a reliable early warning system. This success is directly attributable to the STAT-LSTM's two key architectural features: the Spatio-Temporal Attention mechanism, which enabled it to identify the most

salient input features and time steps across multiple climate variables, and the weighted loss function, which prioritized the accurate prediction of more significant drought events during training. The extensive hyperparameter optimization over 193 trials was crucial in identifying the optimal configuration that maximized these architectural advantages.

Notably, the standard Multivariate LSTM also demonstrated strong performance after hyperparameter tuning across 152 trials, achieving an R^2 of 0.702 and outperforming the statistical baselines. This result validates the potential of deep learning approaches for drought prediction when properly optimized. However, the 18.7% performance gap between the standard Multivariate LSTM and the STAT-LSTM (0.702 vs 0.833 R^2) underscores the critical importance of domain-specific architectural enhancements. The attention mechanism and custom loss function in the STAT-LSTM provided substantial added value beyond standard LSTM architectures, demonstrating that specialized modifications tailored to the drought forecasting task are essential for achieving state-of-the-art performance in climate prediction applications.

B. Comparison with Existing Literature

The results of this study demonstrate that the hybrid STAT-LSTM, with an R^2 of 0.833, is substantially superior to both the statistical baselines (SARIMA and SARIMAX with R^2 of approximately 0.499) and the standard Multivariate LSTM (R^2 of 0.702). The STAT-LSTM's ability to explain over 83% of the variance in drought indices represents a 68% improvement over statistical methods and a 19% improvement over the standard deep learning approach. This finding is consistent with broader trends in environmental forecasting, where many recent studies confirm that hybrid models combining multiple architectural strengths tend to outperform individual component models[6], [11], [14]. This work strongly reinforces the hybrid approach as an effective path forward for climate prediction tasks. This research also builds directly upon existing work for the Vidarbha region. While recent studies have established valuable comparisons between standard ARIMA and LSTM models in this area, this paper takes the critical next step by demonstrating that these common models, while useful, may be insufficient for capturing the full complexity of drought dynamics. The specialized hybrid STAT-LSTM architecture achieves a significant performance gain specific to this region, with the extensive hyperparameter optimization over 193 trials being crucial to unlocking its full potential[12], [13]. Notably, even the standard Multivariate LSTM, when properly tuned over 152 trials, outperformed statistical baselines, validating the potential of deep learning approaches while simultaneously highlighting the substantial added value of domain-specific architectural enhancements.

Furthermore, the success of the STAT-LSTM model validates this architectural choice as a promising direction for drought forecasting research. The work aligns with other advanced, contemporaneous research, such as that by Chen et al. (2025), who also independently developed a model named "STAT-LSTM" for drought prediction[1]. Unlike prior

STAT-LSTM implementations, this study emphasizes extensive hyperparameter optimization, a weighted loss function targeting extreme drought events, and validation on a 20-year region-specific dataset. This convergence suggests a growing consensus that hybrid, attention-based LSTMs with specialized loss functions are highly effective solutions for climate forecasting. The results from both the standard Multivariate LSTM and STAT-LSTM, achieved through rigorous hyperparameter tuning and validation, serve as strong confirmation that this training and testing methodology is highly favorable for capturing the complex spatiotemporal dynamics of drought phenomena.

C. Model Interpretability and Future Directions

While this research provides a strong foundation for the Vidarbha region, it is not without constraints. Currently, the study focuses on a specific geographic area and a fixed set of environmental inputs. Future iterations could refine the STAT-LSTM's accuracy by incorporating higher-fidelity data, such as real-time soil moisture sensors or comprehensive evapotranspiration reanalysis data.

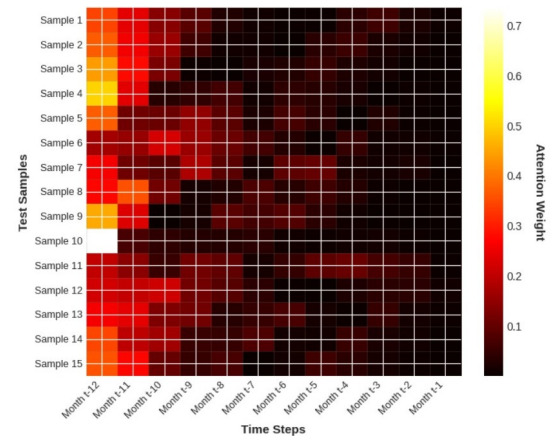


Fig. 4. Attention Heatmap

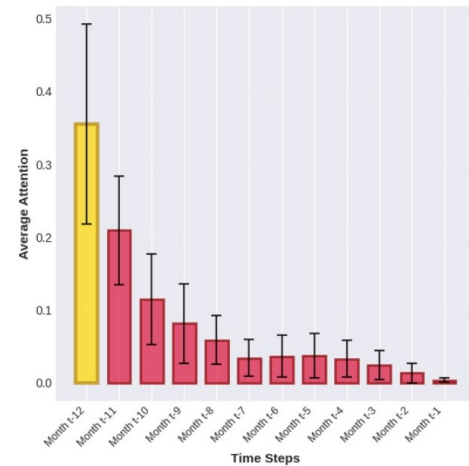


Fig. 5. Average Distribution

A significant milestone in this study was moving beyond “black-box” predictions to understand why the model succeeds. By visualizing the attention weights, direct insights were gained into the model’s decision-making process. As illustrated in Fig. 4 and Fig. 5, the STAT-LSTM demonstrates a clear memory of long-term climatic cycles. The model assigns the highest importance to observations from Month $t-12$, indicating that conditions from exactly one year prior are the strongest predictors of current drought status. This behavior validates the effectiveness of the Spatio-Temporal Attention mechanism in capturing the dominant 12-month seasonal patterns inherent to the region’s climate.

Beyond interpretability, the proposed framework is well-suited for integration into a real-time drought early warning decision system. In an operational setting, near-real-time satellite and meteorological inputs (e.g., rainfall, LST, NDVI) can be continuously ingested, preprocessed using the same normalization and temporal windowing applied during training, and passed through the deployed STAT-LSTM model to generate short-term drought forecasts. The predicted drought indicators can then be mapped against predefined risk thresholds to trigger early warnings, enabling timely interventions for agricultural planning and water resource management. Moving forward, the research can be expanded in the following ways:

- Evaluating the generalization of the STAT-LSTM across drought-prone regions with different climatic regimes and seasonal structures.
- Enhancement in interpretability by analyzing spatial attention weights to identify how specific variables, such as LST or rainfall, influence predictions across seasons.
- Deploying the framework as a real-time early warning system to deliver actionable drought alerts to decision-makers and vulnerable farming communities.

VII. CONCLUSION

This study conducted a rigorous comparative analysis of four forecasting models to identify the most effective method for drought prediction in the Vidarbha region. The models evaluated included statistical baselines (SARIMA and SARIMAX) and deep learning architectures (a standard Multivariate LSTM and a hybrid STAT-LSTM). All deep learning models underwent extensive hyperparameter optimization using the Keras Tuner Hyperband strategy to ensure a fair comparison, with model selection driven strictly by validation performance. The results revealed important insights. The standard Multivariate LSTM, optimized across 152 trials, achieved moderate generalization with a test R^2 of 0.702, outperforming the statistical baselines but exhibiting a larger train-test performance gap. The statistical models, SARIMA and SARIMAX, provided robust baseline forecasts with R^2 values near 0.499 and the lowest RMSE values of 0.290 and 0.291, respectively. The hybrid STAT-LSTM demonstrated superior and more stable generalization, achieving a test R^2 of 0.833 after 193 optimization trials and explaining over 83% of drought variance. Despite the extensive hyperparameter search, the limited divergence between training, validation, and unseen test

performance indicates that the model did not overfit historical data. Instead, its improved predictive capability stems from its domain-informed hybrid architecture, incorporating spatio-temporal attention and a weighted loss function tailored to extreme drought events. These findings confirm that for complex environmental forecasting tasks, carefully constrained, domain-specific hybrid models can substantially outperform both conventional deep learning approaches and traditional statistical methods.

REFERENCES

- [1] Chen, Y., Wu, H., Xie, N. et al. STAT-LSTM: A multivariate spatiotemporal feature aggregation model for SPEI-based drought prediction. *Earth Sci Inform* 18, 289, 2025.
- [2] A. Vaughan et al., “Convolutional Conditional Neural Processes for Local Climate Downscaling,” *Geosci. Model Dev.*, vol. 15, pp. 251–268, 2022.
- [3] C. Chen et al., “GACNPs: Fine-Grained Drought Monitoring Using Remote Sensing Data Based on Conditional Neural Processes,” *IEEE Geoscience and Remote Sensing Letters*, 2024.
- [4] H. Chen et al., “A Novel Agricultural Drought Index Based on Multi-Source Remote Sensing Data and Interpretable ML (IMLDI),” *Agric. Water Manag.*, vol. 308, 2025.
- [5] Aravindh J, Anand R, “A Deep Learning Model for Integrating Landsat-8 and Sentinel 2 Satellite Images to Improve the Spatiotemporal Fusion Network for Drought Monitoring”, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol.19, 2026.
- [6] Yogesh Dhyani, Rahul Jashvantbhai Pandya, “Deep Learning Oriented Satellite Remote Sensing for Drought and Prediction in Agriculture”, 2021 IEEE 18th India Council International Conference (INDICON), pp.1-5, 2021.
- [7] N. Smith et al., “Early Warning of Drought-Induced Vegetation Stress Using Satellite-Based Ecological Indicators,” *Ecological Indicators*, vol. 169, 2024.
- [8] Ankit Paudel, B N Bhavana, Aniketh Anchalia, Anirudh Kakati, Shilpa Chaudhari, “Hybrid Model of Fireworks Algorithm and Deep Learning for Drought Prediction using Satellite Data”, 2023 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT), pp.1-5, 2023.
- [9] Shweta Pokhariyal, N. R. Patel, Ajit Govind, “Machine Learning-Driven Remote Sensing Applications for Agriculture in India—A Systematic Review”, *Agronomy*, vol.13, no.9, pp.2302, 2023.
- [10] Tanriverdi, I., Batmaz, İ. AI-driven U.S. drought prediction using machine learning and deep learning. *Clim Dyn* 63, 249 (2025).
- [11] Tian Wang, Xinjun Tu, Vijay P. Singh, Xiaohong Chen, Kairong Lin, Zonglin Zhou, Drought prediction: Insights from the fusion of LSTM and multi-source factors, *Science of The Total Environment*, vol. 902, 2023.
- [12] Jariwala, K. A., and P. G. Agnihotri. “Comparative analysis of drought modeling and forecasting using soft computing techniques.” *Water Resources Management* 37, no. 15 (2023): 6051-6070.
- [13] Gharehbaghi, A., Ghasemlounia, R., Vaheddoost, B. et al. Development of deep learning approaches for drought forecasting: a comparative study in a cold and semi-arid region. *Earth Sci Inform* 18, 78 (2025).
- [14] Vo, T.Q., Kim, S.H., Nguyen, D.H. et al. LSTM-CM: a hybrid approach for natural drought prediction based on deep learning and climate models. *Stoch Environ Res Risk Assess* 37, 2035–2051 (2023).
- [15] Pande, C.B., Sidek, L.M., Varade, A.M. et al. Forecasting of meteorological drought using ensemble and machine learning models. *Environ Sci Eur* 36, 160 (2024).